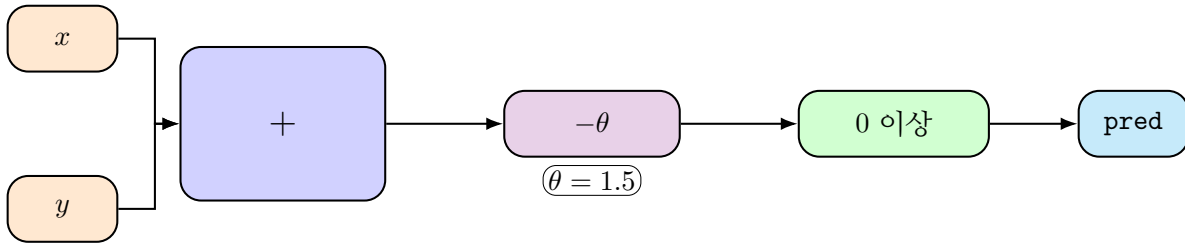


3주차: 오차역전파 학습지

ML기초 시그 by 윤승현

1. 계산 그래프로 보는 AND

$$z = x + y, \quad s = z - \theta \ (\theta = 1.5), \quad \text{pred} = 1[s \geq 0]$$



1. 가능한 모든 입력쌍 $(x, y) \in \{0, 1\}^2$ 에 대해 표를 완성하시오.

x	y	$z = x + y$	$s = z - 1.5$	pred
0	0			
0	1			
1	0			
1	1			

2. $Loss = (pred - true)^2$ 라 하자. $(x, y) = (1, 1)$, $true = 1$ 일 때

$$\frac{\partial Loss}{\partial \theta}$$

를 구하시오.

3.

$$\theta = \theta_{\text{old}} - 0.1 \frac{\partial Loss}{\partial \theta}$$

로 업데이트했을 때 Loss를 구하시오.

4. 위와 같이 업데이트했을 때 Loss가 줄어드는 이유를 간단히 설명하시오.

2. 손으로 훈련시켜봐요(XOR)

$$a = W_1x + b_1, \quad h = \text{ReLU}(a), \quad \hat{y} = W_2h + b_2, \quad \text{Loss} = \frac{1}{2}(\hat{y} - y)^2$$

$$x = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad W_1 = \begin{bmatrix} 1 & 0.5 \\ 0 & 1 \end{bmatrix}, \quad b_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$W_2 = \begin{bmatrix} 1 & 1 \end{bmatrix}, \quad b_2 = 0, \quad y = 0$$

$$\text{ReLU}(t) = \max(0, t)$$

1. 순전파

$$a = \underline{\hspace{2cm}}, \quad h = \underline{\hspace{2cm}}, \quad \hat{y} = \underline{\hspace{2cm}}, \quad \text{Loss} = \underline{\hspace{2cm}}$$

2. 출력층 업데이트

$$\frac{\partial \text{Loss}}{\partial \hat{y}} = \hat{y} - y, \quad \frac{\partial \hat{y}}{\partial W_2} = h, \quad \frac{\partial \hat{y}}{\partial b_2} = 1$$

$$\frac{\partial \text{Loss}}{\partial W_2} = \underline{\hspace{2cm}}, \quad \frac{\partial \text{Loss}}{\partial b_2} = \underline{\hspace{2cm}}$$

$$W_2^{\text{new}} = W_2 - 0.1 \frac{\partial \text{Loss}}{\partial W_2} = \underline{\hspace{2cm}}$$

$$b_2^{\text{new}} = b_2 - 0.1 \frac{\partial \text{Loss}}{\partial b_2} = \underline{\hspace{2cm}}$$

3. ReLU 지나기

$$\text{ReLU}'(t) = \begin{cases} 1 & (t > 0) \\ 0 & (t < 0) \end{cases} \quad \text{ReLU}'(a) = \underline{\hspace{2cm}}$$

$$\frac{\partial \text{Loss}}{\partial a} = \underline{\hspace{2cm}}$$

4. 첫 층 업데이트

$$\frac{\partial \text{Loss}}{\partial W_1} = \underline{\hspace{2cm}}, \quad \frac{\partial \text{Loss}}{\partial b_1} = \underline{\hspace{2cm}}$$

$$W_1^{\text{new}} = W_1 - 0.1 \frac{\partial \text{Loss}}{\partial W_1} = \underline{\hspace{2cm}}$$

$$b_1^{\text{new}} = b_1 - 0.1 \frac{\partial \text{Loss}}{\partial b_1} = \underline{\hspace{2cm}}$$

5. 확인

업데이트 후 다시 순전파하여

$$\hat{y}_{\text{new}} = \underline{\hspace{2cm}}, \quad \text{Loss}_{\text{new}} = \underline{\hspace{2cm}}$$

를 구하시오.